

Denormalization and NoSQL Modelling

Part 1 (Denormalization)

Consider the following normalized relations from a database in a large retail chain:

user(userid, firstName, lastName, email, yearOfBirth, userCategoryId, userZip)
advertiserClient(clientid, clientName, clientContactID, clientZip)
contact(contactid, contactName, contactEmail, contactPhone)
interestArea(userid, interestid, intensity)
interest(interestid, interestLabel)
category(categoryid, categoryName, categoryPriority)
zip(zipcode, city, state)

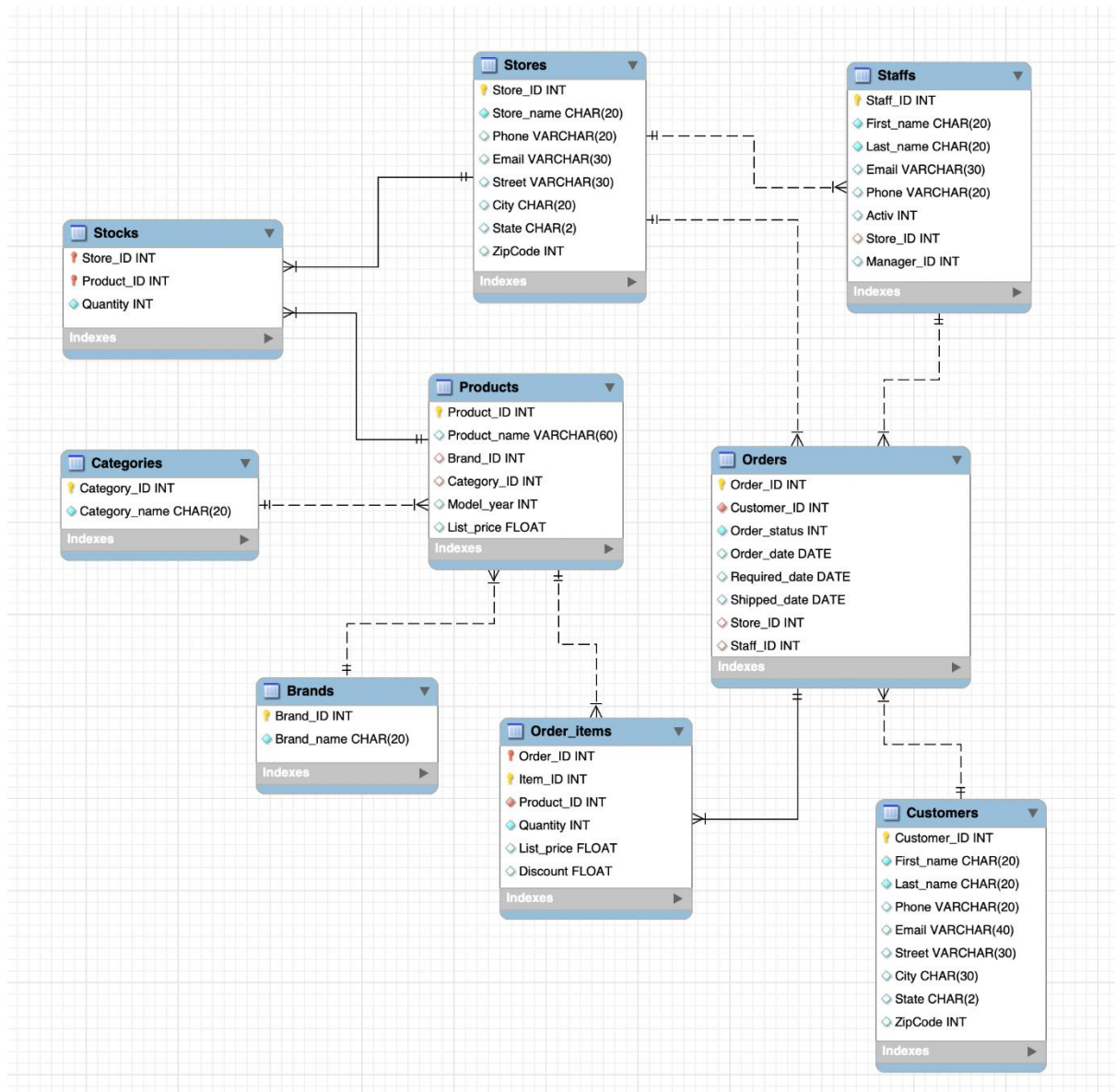
Assume that the mobile service provider has frequent need for the following information:

- List of users sorted by zip code
- Access to a specific client with the client's contact person's name, email, and phone number
- List of users sorted by interest area and within each interest area user's estimated intensity of interest
- List of users within a specific age range sorted by their category and within the category by zip code
- Access to a specific user based on their email address

What opportunities might exist for denormalizing these relations when defining the physical records for this database? Under what circumstances would you consider creating such denormalized records?

Part 2 (Bike Store NoSQL Modelling)

This bike store database tracks sales and production of bike stores in America.



Tasks:

1. Based on the given ERD, state an aggregate boundary
2. Use the visual paradigm to draw aggregate data model for the stated aggregate boundary
3. Use the csv data and the aggregate data model designed in the task 2 to create JSON files.
Each JSON file only contains one JSON object.